WHICH CHORD OF AN EVEN CYCLE PRESERVES ITS SPECTRAL GAP

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ABSTRACT. Add one edge of conductance $w > 0$ to the cycle C_n, $n = 2m \geq 4$, and run the random walk on the resulting weighted graph. We show that the spectral gap of the walk is largest exactly when the new edge joins two antipodal vertices, and that in this case the gap equals $1 - \cos(\pi/m)$, the gap of the cycle itself, for every w. Every other chord strictly lowers the gap, with an explicit bound. If instead the walk is Metropolized so that its stationary law stays uniform, every chord, antipodal or not, strictly lowers the gap. These are one-edge cases of questions of Diaconis about which added edges speed up a random walk most.

1. INTRODUCTION

Diaconis [2, Section 5, Question 4] asks, for a connected regular graph, which perfect matching should be added to make the random walk mix fastest, and, for a general reversible chain, which weighted edge should be added, with the new chain Metropolized to keep its stationary law, to improve mixing or the spectral gap the most. For the cycle with a random perfect matching, the mixing behaviour is now well understood; see Hermon, Sly and Sousi [5] and the references there. It is also known that adding an edge can lower the spectral gap of the random walk; Eldan, Rácz and Schramm [3] show that this happens with substantial probability in random graphs. For the combinatorial Laplacian an added edge never lowers the algebraic connectivity, and the choice of edges that raise it most has been studied by Ghosh and Boyd [4] and by Kim [6]. For a single weighted chord on a weighted cycle, Deng et al. [1] give exact formulas for the effective resistances and the Kirchhoff index after the chord is added, and show that under bounded conductances near-antipodal chords that balance the resistances of the two arcs are near-optimal for the algebraic connectivity.

Here we treat the smallest case completely: a single chord on an even cycle. Write $n = 2m \geq 4$, label the vertices $\mathbb{Z}/n$, give every cycle edge conductance 1, and add one chord $\{0, k\}$, $2 \leq k \leq m$, of conductance $w > 0$ (by rotation and reflection every chord has this form). Let d_i be the weighted degrees, A the weighted adjacency matrix, $P = D^{-1}A$ the random walk, and

$$\gamma = 1 - \lambda_2(P) = \inf_f \frac{\mathcal{E}(f)}{\mathrm{Var}_D(f)}, \qquad \mathcal{E}(f) = \sum_{\{i,j\}} c_{ij}(f_i - f_j)^2, \qquad \mathrm{Var}_D(f) = \min_{c \in \mathbb{R}} \sum_i d_i(f_i - c)^2,$$

where the sum runs over edges, each counted once, and the infimum over nonconstant f. The cycle alone has $\gamma(C_n) = 1 - \cos(2\pi/n)$. Throughout, $a = 1 - \cos(\pi/m) = 1 - \cos(2\pi/n)$.

Theorem 1. *For every $w > 0$, $\gamma(C_{2m} + w\{0, m\}) = a$. For $2 \leq k < m$,*

$$\gamma(C_n + w\{0, k\}) \leq \frac{a}{1 + \frac{2w}{n+w}\cos^2(\pi k/n)} < a. \tag{1}$$

In particular the antipodal chords, and only they, maximise the spectral gap among all single chords of a given weight, and no chord raises the gap of the cycle.

For the second question, take as proposal the walk P above and as target the uniform law. The Metropolis–Hastings kernel is $K(i, j) = P(i, j)\min\{1, P(j, i)/P(i, j)\} = c_{ij}/\max(d_i, d_j)$ for

$i \neq j$, with the diagonal chosen so that rows sum to one. It is symmetric, so the uniform law is stationary.

Theorem 2. *For every chord $\{0,k\}$, $2 \le k \le m$, and every $w > 0$, with $b = w/(2+w)$, $\theta = 2\pi/n$ and $\alpha = \pi k/n$,*

$$1 - \lambda_2(K) \le a - \frac{4b}{n}\big(a^2\cos^2\alpha + \sin^2\theta\,\sin^2\alpha\big) < a.$$

So the Metropolized chain is strictly slower, in the sense of the spectral gap, than the walk on the cycle without the chord, whichever chord is added.

The antipodal case is the point of Theorem 1: the upper bound for the other chords comes from one test function, but the equality needs a lower bound on the whole Rayleigh quotient. The question for full matchings remains open.

2. The random walk

Lemma 3. *If $g_0, \dots, g_m$ are real with $g_0 = g_m = 0$, then $\sum_{i=0}^{m-1}(g_{i+1} - g_i)^2 \ge 2a\sum_{i=1}^{m-1} g_i^2$.*

Proof. The left side is $g^\top T g$ for the $(m-1)\times(m-1)$ tridiagonal matrix T with 2 on the diagonal and -1 next to it. Its eigenvectors are $(\sin(\pi ji/m))_i$, with eigenvalues $2 - 2\cos(\pi j/m)$, $1 \le j \le m-1$, the smallest being $2a$. □

Proof of Theorem 1, antipodal chord. Let $k = m$, so that $d_0 = d_m = 2+w$ and $d_i = 2$ otherwise. The half-turn $(Tf)(i) = f(i+m)$ preserves the graph and the degrees, so it preserves $\mathcal{E}$ and the D-inner product. Write $f = f_+ + f_-$ with $Tf_\pm = \pm f_\pm$. The two parts are orthogonal for both forms, and $\sum_i d_i f_-(i) = 0$, so

$$\mathcal{E}(f) = \mathcal{E}(f_+) + \mathcal{E}(f_-), \qquad \operatorname{Var}_D(f) = \operatorname{Var}_D(f_+) + \sum_i d_i f_-(i)^2.$$

Even part. Put $t = f_+(0) = f_+(m)$ and $g = f_+ - t$. The chord contributes nothing to $\mathcal{E}(f_+)$, and g vanishes at both ends of each of the two arcs from 0 to m. Lemma 3 on both arcs gives $\mathcal{E}(f_+) \ge 2a\sum_i g_i^2$. Since g vanishes where $d_i \neq 2$, $\operatorname{Var}_D(f_+) \le \sum_i d_i g_i^2 = 2\sum_i g_i^2$, so $\mathcal{E}(f_+) \ge a\operatorname{Var}_D(f_+)$.

Odd part. Put $s = f_-(0) = -f_-(m)$. The function f_- has $\sum_i f_-(i) = 0$, so the Poincaré inequality of the cycle, whose Laplacian has second eigenvalue $2 - 2\cos(2\pi/n) = 2a$, gives $\mathcal{E}_{\mathrm{cyc}}(f_-) \ge 2a\sum_i f_-(i)^2$. The chord adds $w(2s)^2$. Since $a \le 1$,

$$\mathcal{E}(f_-) \ge 2a\sum_i f_-(i)^2 + 4ws^2 \ge a\Big(2\sum_i f_-(i)^2 + 2ws^2\Big) = a\sum_i d_i f_-(i)^2.$$

Adding the two parts, $\mathcal{E}(f) \ge a\operatorname{Var}_D(f)$, so $\gamma \ge a$.

For the reverse inequality take $f(i) = \sin(\pi i/m)$. It vanishes at 0 and m, so the chord does not affect $(D-A)f$ and $(D-A)f = 2af = aDf$. Also $\sum_i d_i f(i) = 2\sum_i f(i) = 0$. Hence f is an eigenvector of P with eigenvalue $1-a$ that is orthogonal to the constants, and $\gamma \le a$. □

Proof of Theorem 1, other chords. Let $2 \le k < m$, $\theta = 2\pi/n$, $t = \cos(\pi k/n) > 0$ and $f(i) = \cos(\theta(i - k/2))$. Then $f(0) = f(k) = t$, so the chord contributes no energy, and f is an eigenvector of the cycle Laplacian with eigenvalue $2a$, with $\sum_i f(i) = 0$ and $\sum_i f(i)^2 = n/2$. Hence $\mathcal{E}(f) = na$. With $\sum_i d_i f(i)^2 = n + 2wt^2$, $\sum_i d_i f(i) = 2wt$ and $\sum_i d_i = 2n + 2w$,

$$\operatorname{Var}_D(f) = n + 2wt^2 - \frac{(2wt)^2}{2n+2w} = n\Big(1 + \frac{2wt^2}{n+w}\Big),$$

and $\gamma \le \mathcal{E}(f)/\operatorname{Var}_D(f)$ is (1). □

Remark 4. The same test function gives the bound (1) for every chord of an odd cycle, with $a = 1-\cos(2\pi/n)$, since the argument uses only $\cos(\pi k/n) > 0$, which holds for $2 \le k \le (n-1)/2$.

3. The Metropolized walk

Proof of Theorem 2. The chord endpoints have degree $2 + w$ and every other vertex degree 2. So K has conductance $1/(2 + w)$ on the four cycle edges at the chord endpoints (call this set S), $1/2$ on the other cycle edges, and $w/(2 + w) = b$ on the chord. Since $1/(2 + w) = \frac{1}{2} - \frac{b}{2}$, for every real f,

$$f^\top (I - K)f = \tfrac{1}{2}\mathcal{E}_{\mathrm{cyc}}(f) - \tfrac{b}{2} \sum_{\{i,j\}\in S} (f_i - f_j)^2 + b(f_0 - f_k)^2.$$

The four edges of S are distinct because $2 \le k \le n-2$. Take again $f(i) = \cos(\theta i-\alpha)$, which has mean zero, $\sum_i f(i)^2 = n/2$, $\mathcal{E}_{\mathrm{cyc}}(f) = na$ and $f(0) = f(k) = \cos\alpha$. Its values at the neighbours of the endpoints are $f(\pm 1) = f(k \mp 1)$, equal to $\cos(\theta \mp \alpha)$, and a short computation gives

$$\sum_{\{i,j\}\in S} (f_i - f_j)^2 = 2\big[(a\cos\alpha - \sin\theta\sin\alpha)^2 + (a\cos\alpha + \sin\theta\sin\alpha)^2\big] = 4\big(a^2\cos^2\alpha + \sin^2\theta\sin^2\alpha\big).$$

Dividing $f^\top(I - K)f$ by $\sum_i f(i)^2 = n/2$ gives the bound. The bracket is positive because $a > 0$ and $\sin\theta > 0$ for $n \ge 4$, so the bound is below a, the gap of the walk on C_n. □

Remark 5. Theorem 2 fixes one natural proposal, the walk on the augmented graph. Diaconis's question allows any way of Metropolizing, and among the slower kernels of Theorem 2 we do not determine the best chord.

Remark 6. Exact computations agree with both theorems. For even $n \le 16$, every chord and $w \in \{1/2, 1, 5\}$, the characteristic polynomial of the pencil $(D - A) - xD$ was computed in rational arithmetic in two independent ways, by direct determinants and by eliminating the two path interiors, and the position of its second root was certified by Sturm sequences: 21 equalities and 63 strict inequalities, as predicted. Floating-point checks for even $n \le 40$, five weights from 0.01 to 1000, and every chord, and for odd $n \le 29$, agree with (1) and Theorem 2. The code is in the repository listed below.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [7] and working together inside the Brain framework [9], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [8], the Math Brain repository [7] and the Brain framework repository [9].

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [9] and Math Brain [7]. In September and November 2026 Claude agents (Anthropic) and Codex agents running OpenAI's GPT-6.1, connected to the Brain, studied Diaconis's question. A Codex agent proved both theorems and wrote the exact checks; Claude agents audited the proofs line by line and checked two algebraic steps of the proof of Theorem 2 in Lean, and Claude Opus 5.5 (Anthropic) re-derived the proofs, checked them numerically, and prepared the manuscript. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Code and exact computations are at
`https://github.com/yspennstate/one-chord-even-cycle-spectral-gap`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [7].

References

[1] J. Deng, Liu Chang, Q. Yang, Single-chord augmentation of weighted cycles for algebraic connectivity and network coherence, preprint, 2026, arXiv:2605.24479.

[2] P. Diaconis, Some things we've learned (about Markov chain Monte Carlo), Bernoulli 19 (2013), 1294–1305, doi:10.3150/12-BEJSP09, arXiv:1309.7754.

[3] R. Eldan, M. Z. Rácz, T. Schramm, Braess's paradox for the spectral gap in random graphs and delocalization of eigenvectors, Random Structures Algorithms 50 (2017), 584–611, doi:10.1002/rsa.20696.

[4] A. Ghosh, S. Boyd, Growing well-connected graphs, in: Proc. 45th IEEE Conference on Decision and Control, 2006, 6605–6611, doi:10.1109/CDC.2006.377282.

[5] J. Hermon, A. Sly, P. Sousi, Universality of cutoff for graphs with an added random matching, Ann. Probab. 50 (2022), 203–240, doi:10.1214/21-AOP1532, arXiv:2008.08564.

[6] Y. Kim, Bisection algorithm of increasing algebraic connectivity by adding an edge, IEEE Trans. Automat. Control 55 (2010), 170–174, doi:10.1109/TAC.2009.2033763.

[7] Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, `https://github.com/yspennstate/math-brain-public`.

[8] Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, `https://github.com/yspennstate/math-brain-public/tree/main/papers/math-brain-2026-10-24`.

[9] Y. Shmalo, *Brain Framework*, public software repository, 2026, `https://github.com/yspennstate/brain-framework-public`.

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